M1 · Foundation: innate immunity in context

Innate immunity lecture pack · EAACI/UEMS Knowledge Exam preparation

After this module you can

- Compare innate and adaptive immunity by speed, specificity, receptors, diversity and memory.
- Name the physical, chemical and microbiological layers of the epithelial barrier and their key molecules.
- Place the immediate, early induced and adaptive phases of a response on a time line.
- Link the signs of inflammation to vascular changes and mediators.
- Explain fever, leukocytosis and the acute-phase response, and interpret a raised C-reactive protein (CRP).
- Explain how innate immunity gives the second signal that starts and shapes adaptive immunity.

APR	acute-phase response
CRP	C-reactive protein
CTL	cytotoxic T lymphocyte
DAMP	damage-associated molecular pattern
DC	dendritic cell
DIC	disseminated intravascular coagulation
ICAM-1 / VCAM-1	intercellular / vascular cell adhesion molecule 1
IEL	intraepithelial lymphocyte
IFN	interferon
Ig	immunoglobulin
IL	interleukin
ILC	innate lymphoid cell
LPS	lipopolysaccharide
MBL	mannose-binding lectin
MHC	major histocompatibility complex
NK cell	natural killer cell
PAF	platelet-activating factor
PAMP	pathogen-associated molecular pattern
PGE2	prostaglandin E2
PRR	pattern recognition receptor
PTX3	pentraxin 3
SAP	serum amyloid P
SIRS	systemic inflammatory response syndrome
SP-A / SP-D	surfactant protein A / D
TCR	T-cell receptor
TLR	Toll-like receptor
TNF	tumor necrosis factor
TSLP	thymic stromal lymphopoietin

1. Two branches of one immune system

Innate immunity is always present. It gives continuous barrier defense and reacts to microbes in the first hours or days of an infection, before adaptive responses develop Abbas p.3. The innate system is fully functional before an infection occurs Abbas p.3.

Adaptive immunity needs exposure to the microbe. It increases in magnitude and defensive capability with each repeated exposure Abbas p.3. Lymphocyte activation and differentiation take several days. For this reason the adaptive response is slow, and innate immunity must protect first Abbas p.10.

The two branches work together. The innate response gives early danger signals that start adaptive responses. Adaptive responses, in turn, often make innate mechanisms stronger Abbas p.3.

Key differences

Innate immunity recognizes molecules shared by groups of related microbes and molecules from damaged host cells Abbas p.4. It recognizes only about 1000 such products. Adaptive immunity can recognize millions of antigens Abbas p.63.

Innate immunity uses germline-encoded, invariant receptors. Adaptive immunity uses diverse receptors from somatically rearranged genes Abbas p.63. There are fewer than 100 types of innate receptors. Adaptive immunity has only two receptor types, immunoglobulin (Ig) and the T-cell receptor (TCR), each with millions of variants Abbas p.64. Innate receptors are nonclonal: all cells of one lineage carry the same receptors. Adaptive receptors are clonal Abbas p.64.

Innate memory is limited, while adaptive immunity has memory Abbas p.4. Innate responses to a repeated exposure usually have the same kinetics, magnitude and quality as the first response Abbas p.4. Neither soluble nor cellular innate components generate long-term protective memory Janeway p.37. Neither branch normally reacts against self Abbas p.4.

Innate vs adaptive immunity

Feature	Innate	Adaptive
Speed	Ready before infection; acts in minutes to hours	Days after first exposure; faster and larger on re-exposure
Specificity	Shared patterns: PAMPs, DAMPs (about 1000 structures)	Fine detail of each antigen; millions, also non-microbial
Receptors	Germline-encoded, invariant PRRs; fewer than 100 types	Ig and TCR only, made by somatic recombination
Diversity	Low	Very high
Distribution	Nonclonal: same receptors on all cells of one lineage	Clonal: each clone has its own receptor
Memory	Limited: repeat response is usually the same as the first	Yes: memory cells give a faster, larger 2nd response
Self-reactivity	None: host cells lack ligands or express regulators	Self-reactive cells removed; imperfect (autoimmunity)
Components	Epithelia, phagocytes, DCs, NK cells, mast cells, ILCs, complement	B and T lymphocytes; antibodies

D1. Innate versus adaptive immunity. Innate immunity is fast and broad; adaptive immunity is slow at first, very specific, and has memory. Abbas p.4; Abbas p.63; Abbas p.64; Abbas p.65; Abbas p.6; Abbas p.3; Abbas p.10; Janeway p.38

> **Exam pearl**
>
> "Limited memory" is the textbook term for innate immunity Abbas p.4. The sources do not describe "trained immunity" (innate immune memory). *Not covered in provided sources — needs expert input.*

Components and strategies

Innate components are epithelia and antimicrobial molecules, complement and lectins, and cells: phagocytes, dendritic cells (DCs), natural killer (NK) cells, mast cells, eosinophils and innate lymphoid cells (ILCs). Adaptive components are B and T lymphocytes and antibodies Abbas p.4. DCs, some macrophages and mast cells live in most tissues and act as sentinels Abbas p.4.

Innate immunity has two main strategies Abbas p.4:

- Inflammation: recruitment of phagocytes and other leukocytes that destroy microbes.
- Antiviral defense: blocking viral replication or killing virus-infected cells.

A third function is to stimulate adaptive immunity Abbas p.89. Defects in innate components are very rare. They increase susceptibility to infection even when adaptive immunity is intact Janeway p.37.

What innate immunity recognizes

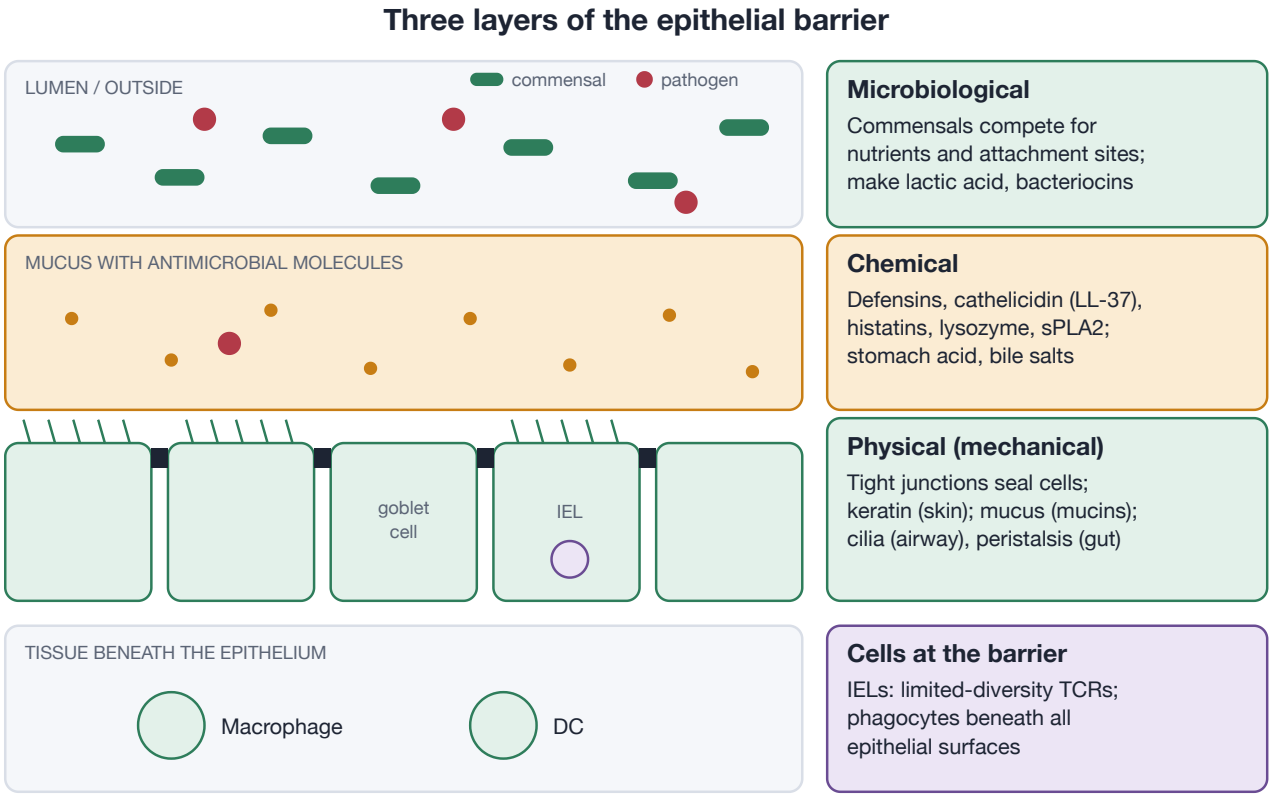
Microbial structures that stimulate innate immunity are often shared by classes of microbes. They are called pathogen-associated molecular patterns (PAMPs) Abbas p.63. These products are often essential for microbial survival. Therefore microbes usually cannot evade innate immunity by losing them Abbas p.63.

Damaged or dying host cells release damage-associated molecular patterns (DAMPs). DAMPs also signal sterile injury, for example burns or trauma Abbas p.63. Cells that die by apoptosis generally do not release DAMPs Abbas p.63. Some cytokines released from damaged cells act as DAMPs and are called alarmins Abbas p.63.

Innate immunity does not attack healthy cells. Normal cells lack most ligands, the receptors sit in separate cell compartments, and regulatory proteins on host cells block activation Abbas p.65. Pattern recognition receptors (PRRs), including Toll-like receptors (TLRs), are the subject of M3.

2. Epithelial barriers: the first line of defense

The main interfaces with the environment are the skin and the mucosa of the gastrointestinal, respiratory and genitourinary tracts Abbas p.79. Surface epithelia give three types of barrier: mechanical (physical), chemical and microbiological Janeway p.43. Phagocytes lie beneath all epithelial surfaces Janeway p.38.



D2. Barrier layers of a mucosal surface (schematic, not to scale). Black bars between cells are tight junctions. IEL, intraepithelial lymphocyte; sPLA2, secretory phospholipase A2.

Janeway p.43; Abbas p.80; Janeway p.45; Janeway p.44; Janeway p.38; Janeway p.48

2.1 Physical (mechanical) barrier

Epithelial cells form tight junctions. Tight junctions block the passage of microbes between the cells Abbas p.80. In the skin, the outer keratin layer blocks microbial penetration. Skin infections usually occur after a barrier breach Abbas p.80. Loss of epithelial integrity predisposes to infection. Infection of barrier cells is often the first step in barrier breakdown Abbas p.79-80.

Respiratory, gastrointestinal and genitourinary epithelia produce mucus. Mucus contains glycoproteins called mucins and physically impairs microbial invasion Abbas p.80. Ciliary action in the bronchial tree and peristalsis in the gut remove microbes Abbas p.80.

> **Clinical correlation**
>
> - Cystic fibrosis: thick mucus or impaired ciliary movement leads to lung infections. The bacteria colonize the epithelial surface but do not cross it Janeway p.44.
> - Failure of peristalsis: typically followed by overgrowth of pathogenic bacteria in the gut lumen Janeway p.44.

2.2 Chemical barrier

In the upper gastrointestinal tract, stomach acid, digestive enzymes, bile salts, fatty acids and lysolipids form a chemical barrier Janeway p.44. Epithelial cells and phagocytes also secrete antimicrobial enzymes and peptides.

Molecule	Main source	Key facts
Lysozyme	Tears, saliva, phagocytes, Paneth cells	Glycosidase that cleaves peptidoglycan. More effective against Gram-positive bacteria, because LPS covers the peptidoglycan of Gram-negative bacteria Janeway p.45.
Secretory phospholipase A2	Tears, saliva, phagocytes, Paneth cells	Antibacterial enzyme secreted with lysozyme Janeway p.45.
Defensins (alpha, beta)	Mucosal epithelia; neutrophils, NK cells, CTLs	Cationic, hydrophobic peptides of 29–34 amino acids with three disulfide bonds Abbas p.80. Disrupt membranes of bacteria, fungi and some enveloped viruses within minutes Janeway p.45.
Alpha-defensins	Paneth cells (small-bowel crypts)	Paneth cells are a major source Abbas p.80.
Beta-defensins	Airway, urogenital tract, skin, tongue	Generally made in response to microbial products Janeway p.46.
Cathelicidin (LL-37)	Neutrophils; skin, gut and airway epithelia	18-kD precursor, cleaved into two active peptides Abbas p.80. LL-37 binds and neutralizes LPS Abbas p.80.
Histatins	Parotid, sublingual, submandibular glands	Act against *Cryptococcus neoformans* and *Candida albicans* Janeway p.46.

Defensins, cathelicidins and histatins are the three important classes of antimicrobial peptides in mammals Janeway p.45. Most antimicrobial peptides are made as inactive proproteins. A protease must cleave them to activate them Janeway p.47. In neutrophils, cathelicidin propeptides sit in secondary granules. Elastase from primary granules activates them in the phagosome Janeway p.46.

> **Exam pearl**
>
> - Abbas describes two human defensin families, alpha and beta Abbas p.80. Janeway also names theta-defensins, but the single human theta-defensin gene is inactivated Janeway p.46.
> - DNA-bound LL-37 activates plasmacytoid DCs to secrete type I IFNs. LL-37 may be an autoantigen in psoriasis Abbas p.80.

Barrier epithelia also contain intraepithelial T lymphocytes. Their antigen receptors have limited diversity Abbas p.80.

2.3 Microbiological barrier: the commensal microbiota

1. Commensals compete with pathogens for nutrients and attachment sites Janeway p.44.
2. Commensals make antimicrobial substances, for example lactic acid from vaginal lactobacilli, and bacteriocins Janeway p.44.
3. Commensals stimulate epithelial cells to make antimicrobial peptides Janeway p.44.

> **Clinical correlation**
>
> Antibiotics that kill commensals often let pathogens replace them and cause disease Janeway p.44. Commensals themselves can cause disease if their growth is not controlled or the host is immunocompromised Janeway p.44.

> **Analogy**
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> Think of a castle. The wall and moat are the physical barrier. Boiling oil poured from the wall is the chemical barrier. Friendly tenants who occupy every room leave no space for intruders: that is the microbiota.

3. Time course of the response to infection

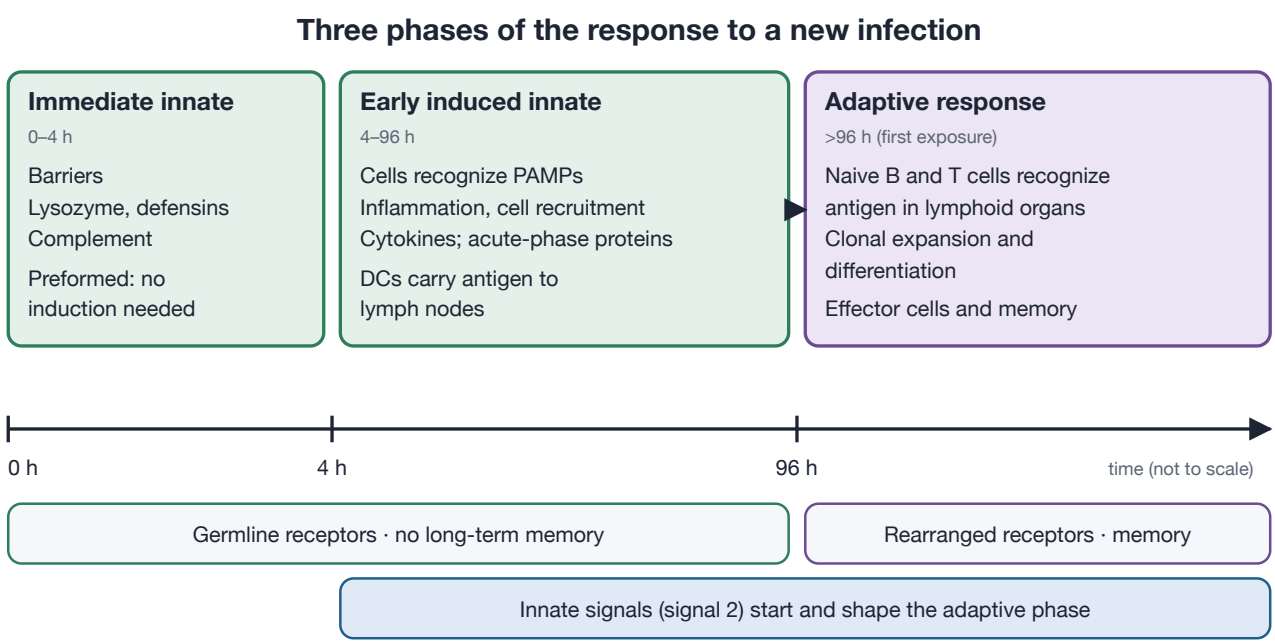
The response to a new pathogen has three phases Janeway p.37:

1. **Immediate innate phase (0–4 h).** Preformed soluble molecules act at once: lysozyme, defensins and complement Janeway p.37.
2. **Early induced innate phase (4–96 h).** Innate cells recognize PAMPs and become activated Janeway p.37. An inflammatory response often first counters the spread of the pathogen Janeway p.42.

Induced responses act over several days. During this time DCs carry antigen to local lymphoid tissue Janeway p.43.

3. **Adaptive phase (>96 h).** Rare specific B and T cells must first expand clonally. The effector mechanisms are similar in each phase Janeway p.37.

Innate defenses start within minutes. An adaptive response is necessary only when innate defenses are bypassed, evaded or overwhelmed Janeway p.38. Even then, innate immunity keeps pathogen numbers in check while the adaptive response develops Janeway p.38.



D3. Time course of the response to a first infection. Phase boundaries are approximate and vary between infections.
Janeway p.37; Janeway p.42; Janeway p.43; Janeway p.110; Abbas p.3; Abbas p.100

4. Inflammation

Acute inflammation is the accumulation of leukocytes, plasma proteins and fluid from the blood at an extravascular site of infection or injury Abbas p.89. Tissue sentinel cells (macrophages, DCs, mast cells) and epithelial cells start it after they recognize PAMPs or DAMPs Abbas p.90.

Inflammation has three roles Janeway p.82-83:

- Deliver more effector molecules and cells from the blood to the site of infection.
- Induce local blood clotting, a physical barrier to spread through the blood.
- Promote repair of injured tissue.

4.1 Cardinal features and their causes

Inflammation is characterized by pain, redness, heat and swelling at the site of infection Janeway p.83. Each sign reflects a change in the local blood vessels.

Vessel change	Clinical sign / effect	Source
Vasodilation, increased local blood flow, slower flow along vessel walls	Heat and redness	Janeway p.83
Endothelial activation with adhesion molecules	Leukocytes attach and migrate into tissue (extravasation)	Janeway p.83
Increased vascular permeability	Swelling (edema) and pain; complement and MBL enter tissue	Janeway p.83
Clotting in microvessels	Pathogen cannot spread through the blood	Janeway p.83

The sources list four signs. They do not list "loss of function" as a fifth cardinal sign. *Not covered in provided sources — needs expert input.*

4.2 Mediators

- **Histamine and prostaglandins** from mast cells cause vasodilation and increased capillary permeability Abbas p.94.
- **Lipid mediators** from macrophages and neutrophils: prostaglandins, leukotrienes and platelet-activating factor (PAF) Janeway p.83.
- **Bradykinin.** Tissue injury activates the kinin system. Bradykinin increases vascular permeability and causes pain Janeway p.84.
- **C5a** increases vascular permeability. It also makes mast cells release histamine and TNF Janeway p.84.
- **Cytokines.** TNF, IL-1 and IL-6 are three of the most important proinflammatory cytokines. Macrophages and DCs make most innate cytokines Abbas p.90-91.

4.3 Sequence of events in acute inflammation

1. Sentinel cells recognize PAMPs or DAMPs Abbas p.90.
2. Mast cells release histamine and prostaglandins. Vessels dilate and become leaky Abbas p.94.
3. TNF and IL-1 induce E-selectin, ICAM-1 and VCAM-1 on endothelium and stimulate chemokines such as CXCL8 and CCL2 Abbas p.94.
4. Neutrophils arrive first. They are the most abundant leukocytes in blood and respond most rapidly to chemotactic signals Abbas p.94.
5. Monocytes become more numerous over time. They may dominate some reactions, because they live longer and can proliferate locally Abbas p.94.

Acute inflammation develops in minutes to hours and lasts for days Abbas p.94. Chronic inflammation follows if infection or injury persists. It involves monocytes and lymphocytes, angiogenesis and fibrosis Abbas p.95. Leukocyte migration and phagocytosis are the subject of M4.

Allergy link

Barrier epithelial cells make IL-25, thymic stromal lymphopoietin (TSLP) and IL-33. These cytokines stimulate ILC2s, Th2 cells and mast cells and contribute to allergic disease. IL-33 is an alarmin: damaged epithelial cells release it rapidly Abbas p.94.

5. Systemic effects: fever, leukocytosis and the acute-phase response

5.1 Fever

1. TNF, IL-1 and IL-6 are endogenous pyrogens Abbas p.96; Janeway p.109.
2. They induce cyclooxygenase-2, which makes prostaglandin E2 (PGE2) Janeway p.109.
3. PGE2 acts on the hypothalamus. Heat production increases and vasoconstriction reduces heat loss Janeway p.109.

Lipopolysaccharide (LPS) is an exogenous pyrogen. It also induces cyclooxygenase-2 directly through TLR4 Janeway p.109. Prostaglandin synthesis inhibitors such as aspirin reduce fever Abbas p.97.

Source difference

Abbas states that the role of fever in host defense is not well understood Abbas p.97. Janeway states that fever is generally beneficial: most pathogens grow better at lower temperatures, and adaptive responses are more intense at higher temperatures Janeway p.109. Abbas is the preferred source for this pack.

5.2 Leukocytosis

TNF, IL-1 and IL-6 reach the bone marrow. Together with colony-stimulating factors they increase blood neutrophils and monocytes (leukocytosis) Abbas p.97.

5.3 Acute-phase response

TNF, IL-1 and IL-6 make hepatocytes produce acute-phase proteins, for example CRP, serum amyloid P (SAP) and fibrinogen. A raised plasma CRP is a clinical sign of inflammation Abbas p.97. In infection, CRP can rise up to 1000-fold Abbas p.88. Blood levels of some other proteins go down Janeway p.109.

Acute-phase protein	Function
CRP (short pentraxin) Abbas p.87	Binds phosphocholine on bacteria and fungi, but not in host membranes. Opsonizes and binds C1q Janeway p.110; Abbas p.88.
SAP (short pentraxin)	Binds C1q and activates the classical pathway Abbas p.88.
PTX3 (long pentraxin)	Made by DCs, macrophages and endothelium; also stored in neutrophil granules Abbas p.88. Binds C1q Abbas p.88.
MBL (collectin)	Low in health; increases in the acute-phase response Janeway p.110.
SP-A, SP-D (collectins)	Increase in the acute-phase response; promote phagocytosis of *Pneumocystis jirovecii* Janeway p.110.
Fibrinogen	Acute-phase protein made by hepatocytes Abbas p.97.

Within a day or two, acute-phase proteins give antibody-like functions. They have no structural diversity, and any stimulus that releases TNF, IL-1 and IL-6 induces them Janeway p.110.

> **Source difference**
>
> Abbas lists SAP as a human acute-phase protein Abbas p.97. Janeway lists SAP as an acute-phase protein in mice but not in humans Janeway p.111. Follow Abbas, but know that the two books differ.

5.4 When the systemic response is harmful

Sepsis shows fever, fast heart and respiratory rates, metabolic abnormalities and mental disturbance. LPS or lipoteichoic acid often starts it Abbas p.97. In septic shock, high TNF causes vascular collapse and disseminated intravascular coagulation (DIC) Abbas p.97. Systemic TNF reduces myocardial contractility and vascular tone, causes intravascular thrombosis and, if prolonged, cachexia Abbas p.97. Anti-TNF antibodies and soluble TNF receptors have not shown benefit in clinical sepsis trials Abbas p.97. A similar syndrome after burns, trauma or pancreatitis is the systemic inflammatory response syndrome (SIRS) Abbas p.97.

> **Clinical correlation**
>
> Too little TNF, for example with TNF-blocking drugs, may cause failure to contain infections Abbas p.94. TNF blockade can reactivate tuberculosis in apparently well patients with previous infection Janeway p.109.

6. Innate immunity instructs adaptive immunity

Lymphocyte activation needs two signals Abbas p.100:

1. **Signal 1** is antigen. It makes the response specific Abbas p.100.

2 **Signal 2** comes from the innate response to microbes or injured cells. It makes sure that the adaptive response starts for a dangerous infection, not for harmless or self antigens Abbas p.100.

Second signals are costimulators (for T cells), cytokines (for T and B cells) and complement breakdown products (for B cells) Abbas p.100. They increase the magnitude of the adaptive response and also influence its type Abbas p.100.

6.1 Dendritic cells: the bridge

PAMP-activated DCs express costimulators. They direct naive T cells to effector types such as Th1 or Th17 Abbas p.81. TNF stimulates DC migration to the lymph node and maturation into highly costimulatory antigen-presenting cells Janeway p.111.

6.2 Innate cytokines shape the T-cell response

Innate cytokine(s)	Effect on adaptive immunity
IL-12	Naive CD4+ T cells → Th1
IL-1, IL-6, IL-23	Naive CD4+ T cells → Th17
IL-25, IL-33, TSLP	Naive CD4+ T cells → Th2 (allergy-relevant)
IL-15	Survival of memory CD8+ T cells
IL-6	Survival of antibody-producing plasma cells

Table source: Abbas p.101

Blood-borne microbes activate the alternative complement pathway. Complement products then enhance antibody production by B cells Abbas p.101. The link also works in the other direction: IgG antibodies coat microbes and engage phagocytes Abbas p.95.

6.3 Adjuvants use this principle

Most adjuvants work by stimulating innate responses at the site of antigen exposure. Aluminium salts are the most common adjuvant in human vaccines. They may act through inflammasome activation Abbas p.101. CpG, a TLR9 ligand, is used in some hepatitis B virus vaccines Abbas p.101. Adjuvants make DCs express more MHC molecules (signal 1), costimulators (signal 2) and cytokines, and migrate to lymph nodes Abbas p.101.

> **Analogy**
>
> Signal 1 is the key that fits only one lock. Signal 2 is the alarm that says "this is a real emergency". The car starts only with both.

6.4 Brakes on innate responses

IL-10 is made by macrophages and DCs and inhibits them: a negative-feedback loop. Loss-of-function mutations in the IL-10 receptor cause severe colitis in infancy Abbas p.101.

7. Roadmap of the pack

- M2 Cellular components: neutrophils, macrophages, DCs, NK cells, mast cells, eosinophils, basophils, ILCs.
- M3 PRRs & signalling: TLRs, NOD-like receptors and the inflammasome, cytosolic nucleic-acid sensors, soluble PRRs.
- M4 Phagocytosis & phagocyte defects: extravasation, respiratory burst, neutrophil extracellular trap (NET), chronic granulomatous disease (CGD), leukocyte adhesion deficiency (LAD).
- M5 Complement: three pathways, C3 convertase, C5 convertase, membrane attack complex (MAC), regulators.
- M6 Complement deficiencies & HAE: deficiency phenotypes, laboratory work-up, hereditary angioedema (HAE) and C1 inhibitor (C1-INH).

5 key points

1. Innate immunity is preformed, fast and broad. It uses fewer than 100 germline-encoded, nonclonal receptors and has limited memory Abbas p.64; Abbas p.4.
2. Epithelia give mechanical, chemical and microbiological barriers. Key chemical molecules are defensins, cathelicidin (LL-37), histatins and lysozyme Janeway p.43; Janeway p.45.
3. The response has three phases: immediate innate 0–4 h, early induced innate 4–96 h, adaptive >96 h Janeway p.37.
4. TNF, IL-1 and IL-6 drive local inflammation and the systemic effects: fever, leukocytosis and hepatic acute-phase proteins such as CRP Abbas p.91; Abbas p.97.
5. Innate immunity supplies signal 2 (costimulators, cytokines, complement fragments), which starts and shapes adaptive immunity Abbas p.100.

3 exam traps

1. Lysozyme is more effective against Gram-positive bacteria, not Gram-negative bacteria. LPS covers the peptidoglycan of Gram-negative bacteria Janeway p.45.
2. Acute-phase proteins act like antibodies, but they come from the liver, have no structural diversity and are induced nonspecifically Janeway p.110.
3. Anti-TNF drugs help in chronic inflammatory disease, but they did not help in clinical sepsis trials and can reactivate tuberculosis Abbas p.97; Janeway p.109.

Practice: M1 questions in the quiz (IDs M1-A1 … M1-K2).

Sources: Abbas, Cellular and Molecular Immunology, 11th ed., pp. 3–10, 63–65, 79–81, 88–101 · Janeway's Immunobiology, 8th ed., pp. 37–47, 82–84, 109–111 · Fact IDs refer to `ledger/facts.csv`.